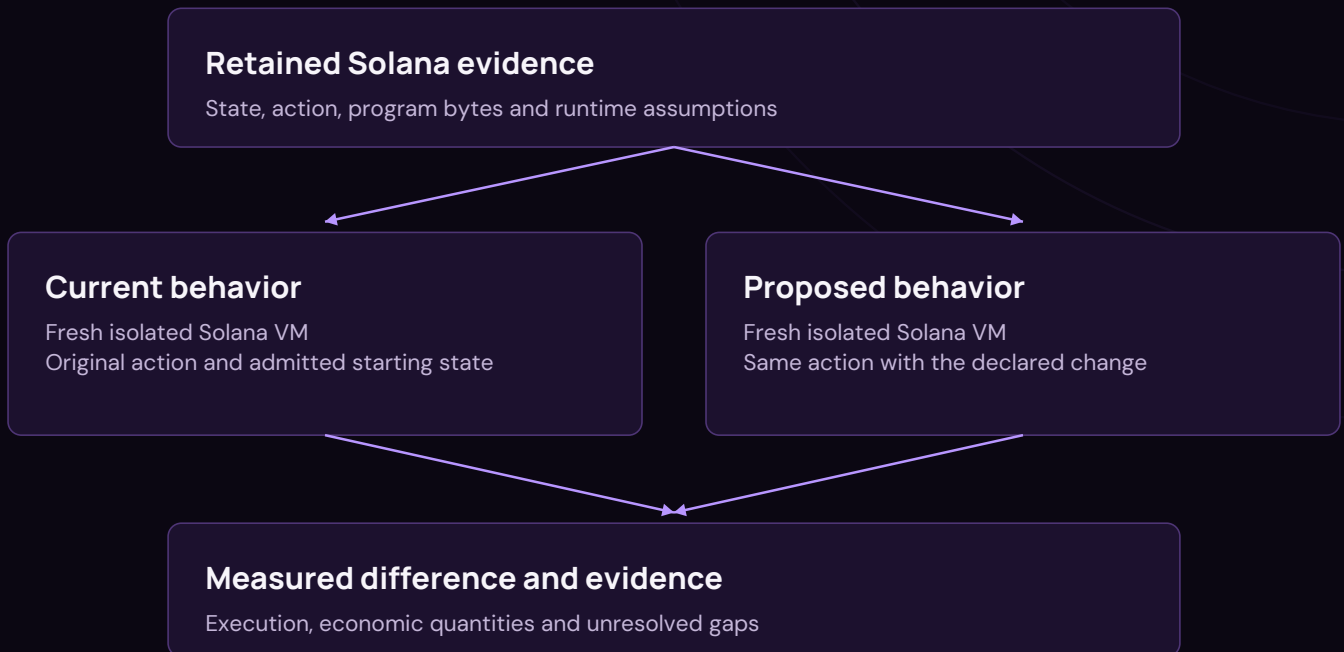


Eplyx Technical Overview

Eplyx measures the consequences of supported on-chain changes before deployment. It reconstructs bounded execution environments from retained Solana evidence, runs current and proposed behavior, and reports behavioral and economic differences.

For protocol developers, security reviewers and governance participants, the useful question is concrete: will the same action still succeed, and what will the user receive? Eplyx makes the inputs, measured consequences and evidence gaps inspectable.



What has been built

The source implements upgrade review, two fee operations, migration rehearsal, lifecycle analysis, bounded code-and-fee comparison, migration ordering, amount search and offline reproduction. Each has a specific evidence contract. [2-11]

Retained examples include an upstream Stake Pool candidate preserving 10 of 10 evaluated transactions, a Token-2022 fee proposal reducing recipient credit by 150 raw units, and 64 search cases executed in 128 VM runs. These are measured examples with explicit scope. [3,5,7]

Source reviewed on 4 October 2026 · main at 08b103f · latest commit dated 2 October 2026

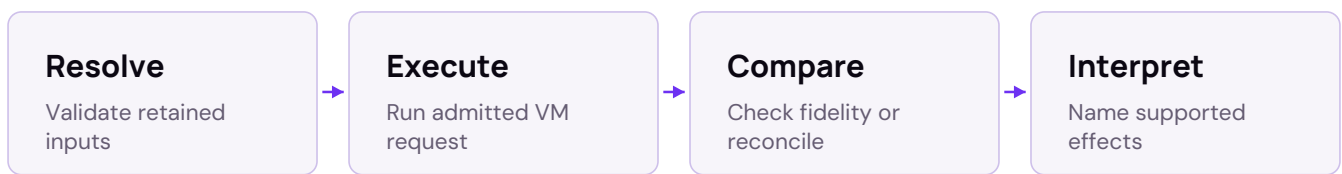
HOW A RESULT IS ESTABLISHED

Execution architecture

The Rust engine owns evidence validation, execution requests, reconciliation and interpretation. LiteSVM 0.16 supplies Solana program execution. The browser presents engine output; it does not calculate protocol economics. [1,11]

Retain and resolve the execution inputs

Historical replay binds the original native transaction, account bytes and presence, program ELF binaries, loaders and dependencies, and runtime evidence. Versioned manifests and SHA-256 content references are checked before execution. Native v0 messages resolve address lookup tables from retained evidence; the newer LegacyV2 envelope binds native legacy messages to the frozen validator transaction. Resolution does not repair missing inputs from the network. [1]



Establish the baseline before interpreting a candidate

Historical upgrade review requires the baseline to reproduce under its declared fidelity profile. Schema 1 preserves its original outcome, fee, CPI-summary and watched-state checks. Schema 2 also checks logs, detailed CPI, return data, account presence and complete watched-account state, plus declared compute usage. Checkpointed profiles validate start and terminal evidence and transaction closure; causal replay can derive a target boundary from preceding transactions. [1]

Current-state fee comparisons use retained-capture admission and independent VM reconciliation. They do not establish a historical validator replay. Both sides use fresh environments and the same action, code and Clock, except for the declared configuration difference. Baseline post-state never becomes proposed pre-state. [3]

Separate execution from economic meaning

Protocol adapters evaluate only their admitted subjects after historical fidelity succeeds. Replay can match without economic coverage. Unexplained account changes remain visible, and semantic provenance records whether an interpretation relies on a standard interface, corroborated execution or repository source. A matched replay alone does not establish source-to-ELF equivalence. [1,13]

Hosted jobs use child workers with empty environments. The filesystem registry and content store hold authoritative analytical bytes; optional PostgreSQL holds mutable identity and authorization. Provider acquisition belongs to the parent service. [11]

CURRENT SOURCE AND SUPPORTED SURFACES

Implemented capabilities

Four typed ChangeSpec kinds identify proposals independently from runs and results.
Configuration analysis currently admits exactly two fee operations. [2,3]

Capability	Implemented boundary	Entry
Program upgrade	Pinned historical baseline versus candidate ELF; behavioral and named economic changes over admitted observations.	CLI and hosted CI
Token-2022 fee	Already-active newer transfer-fee bps; one retained original-owner TransferChecked. [3]	CLI, API and guided form
Stake Pool fee	SOL deposit fee; simulated SetFee followed by one qualified retained DepositSol. [4]	CLI and hosted API
Token migration	Declared conversion terms, actual SBF rehearsal, reconciliation, stress, search and selected gates. [9]	CLI and prepared hosted form
Lifecycle change	Declared policy consequences over immutable snapshots; separate readiness questions. [10]	CLI and prepared hosted form
Code and fee	Independent V1/V2 × current/proposed fee cells for a qualified Stake Pool profile. [6]	CLI and hosted child run
Migration order	A alone, B alone and both orders for two reserve-funded units under one proposal. [9]	CLI and hosted child run
Amount search	One Token-2022 transfer-amount interval, one loss predicate, bounded paired execution. [7]	Local CLI
Selected cases	Two or three distinct retained source accounts, with independent paired results. [8]	Local CLI

Adjacent analytical and governance paths

Current-state paths support bounded transfer, Meteora exit and position-withdrawal checks under their own evidence requirements. A successful path cannot supply proof for another path. Squads V4 records bind a reviewed upgrade to a proposal and buffer, with a separate deployment-attestation contract. They neither sign nor execute proposals. [10,12]

A surface being implemented in source does not attest a particular remote deployment. Search and case-set tools have no hosted form; Stake Pool fee analysis has no standalone guided fee form. [4,7,8,11]

RETAINED QUALIFICATION EXAMPLES

Measured consequences

Token 2022 transfer fee

An untouched finalized capture from 18 September 2026 has a 50 bps newer fee schedule already active at captured epoch 1037. The proposal changes only that rate to 200 bps. The same deployed code and 10,000-raw-unit transfer execute independently. [3]

Measured quantity in raw units	50 bps	200 bps
Source debit	10,000	10,000
Recipient public credit	9,950	9,800
Destination withheld increment	50	200

Recipient credit falls by 150 raw units. Source debit equals recipient credit plus withheld increment on both sides. The proposed mint is a typed, parent-bound derivation; this operation does not execute SetTransferFee or establish fee-authority possession.

Stake Pool SOL deposit fee

For one retained deposit at slot 447850493, a proposal changes the fee from 0/1000 to 1/100. The qualified historical program executes SetFee with an explicitly assumed manager signer. Its verified pool output is the only account substituted into the proposed deposit environment. [4]

Measured quantity in raw pool tokens	Current fee	Proposed fee
Recipient account credit	760,985,008	753,375,157
Independent manager fee credit	0	7,609,851

Both deposits reconcile, with the same 760,985,008 raw mint-supply increase. Recipient and referral roles alias, so no independent referral split is asserted. The schema-1 Clock retains epoch zero and Matched fidelity; it does not reconstruct the slot-accurate validator runtime.

An unmodified upstream candidate

An unmodified program@v2.1.1 source build preserves success, watched state, fees, economics and CPI shapes on all ten records; the gate exits 0 and baseline fidelity is 10/10 Matched. A constructed DepositSol-only regression yields one deposit decrease and nine withdrawal reverts. Neither proves deployment safety or deployed-byte equivalence. [5]

EXPERIMENTS WITH EXPLICIT STATE RELATIONSHIPS

Interactions and order effects

Does the fee effect change after a code change

The interaction analyzer evaluates four action cells in fresh banks. Two separate configuration executions produce the proposed pool state under V1 and V2; only each verified pool output feeds its corresponding proposed-fee action. The admission profile permits the historical same-code control or one constructed configuration-capable fixture. [6]

Recipient credit in raw units	Current fee C0	Proposed fee C1
Historical code V1	R00 760,985,008	R01 753,375,157
Qualified fixture V2	R10 760,985,008	R11 753,375,157

The fee effect is -7,609,851 raw under both versions. Interaction is $(R11 - R10) - (R01 - R00) = 0$. Zero interaction means the measured fee effect is equal for this quantity and case; the fee effect itself is nonzero. Missing or unreconciled cells remain unavailable, rather than becoming zero.

Both ChangeSpec IDs stay independent. This experiment overlays code in local VMs; it does not execute a loader Upgrade, establish rollout order or qualify arbitrary candidate binaries.

Which holder succeeds when reserve is shared

The migration-order fixture quotes A at 50 destination units and B at 60, with an initial reserve of 80. Both solo controls succeed. In A then B, A consumes 50 and B rejects with 30 remaining. In B then A, B consumes 60 and A rejects with 20 remaining. A reserve of 110 is the sufficient-funding control. [9]

Scenario	First result	Second result
A then B	A succeeds; reserve 30	B rejects; rollback checked
B then A	B succeeds; reserve 20	A rejects; rollback checked

Only transactions within an order share the actual VM state. The portable artifact retains initial, intermediate and final account states, known absence, messages, runtime, candidate and reconciliation. It demonstrates a synthetic two-unit ordering effect under one unchanged proposal and fixed Clock. It establishes neither production fairness nor atomic execution.

BOUNDED EXPLORATION AND INDEPENDENT REVIEW

Search and reproducible evidence

Search the action amount without changing observed state

The Token-2022 search varies a hypothetical TransferChecked amount within retained source balance. It reuses paired execution, keeping observed accounts, code, Clock and fee proposal fixed. The declared condition is recipient loss strictly greater than 100 raw units. [7]

Reference search over amounts 1 to 10,000	Retained result
Distinct executed amounts / VM calls	64 / 128
Refinement evaluations / matching cases	16 / 25
Amounts left untested	9,936
Smallest tested matching amount	6,732 raw
Recipient credits at that amount	6,698 to 6,597 raw
Completion	Budget exhausted

The 101-raw-unit loss is a reproducible witness for the selected condition. Sampling and fee-boundary hints do not certify a global minimum or exhaustive coverage. A fee impact is not automatically a vulnerability.

Selected observed cases retain their own contexts

Two retained sources under 50 to 200 bps yield credits of 17,532 to 17,268 and 307,592 to 302,955 raw. Each keeps its own amount, state and Clock. Four fresh VMs establish two reconciled pairs, without population or holder-identity conclusions. [8]

Verification and reproduction answer different questions

Read-only verification checks bytes, identities, derivations and reconciliation without a VM. Reproduction executes again and requires canonical output equality. Hash consistency does not authenticate provider claims or prove a VM ran. Reproduction needs a compatible build and complete artifact. [6-9,11]

```
eplx parameter verify-search --artifact ./search --format json
eplx parameter reproduce-witness --artifact ./search \
--witness SAVED_WITNESS_SHA256 --format json
```

Private reference inputs are separate from public Git. Use the saved artifact and witness identity. [11]

IMMUTABLE REFERENCES FOR TECHNICAL REVIEWERS

Sources and review provenance

All Eplyx source links below are pinned to the reviewed main revision. Dated phase documents record the scope at the time of qualification; the current source and later integration records determine which capabilities are present now.

[1] Execution architecture and evidence policy

[Architecture](#) · [Fidelity implementation](#) · [Native legacy envelope](#)

[2] Proposal kinds and current entry points

[ChangeSpec source](#) · [Getting started](#)

[3] Token 2022 active fee comparison

[Parameter contract](#) · [Qualification JSON](#)

[4] Stake Pool deposit fee comparison

[Qualification JSON](#) · [Fee guide](#)

[5] Upstream candidate and controlled regression

[External candidate case](#) · [Canonical regression guide](#) · [Upstream release](#)

[6] Code and fee interaction

[Interaction contract](#) · [Qualified fixture matrix](#)

[7] Bounded amount search

[Search contract](#) · [Qualification JSON](#)

[8] Selected observed transfer cases

[Case set contract](#) · [Qualification JSON](#)

[9] Token migration and ordering

[Migration guide](#) · [Order case contract](#)

[10] Lifecycle and current state checks

[Lifecycle guide](#) · [Current state guide](#)

[11] Hosted storage and offline execution

[Cloud architecture](#) · [Integration record](#) · [Worker source](#)

[12] Squads governance evidence

[Proposal binding](#) · [Deployment attestation](#)

[13] Historical replay qualification limits

[Phoenix runtime evidence gap](#) · [Drift bounded semantics](#) · [Orca semantic provenance](#)

Reviewed revision 08b103f0a12ac201f7634fa2803f8ea02832992c · 4 October 2026. Technical claims are grounded in current source and retained project qualification evidence. This overview includes no private capture bytes or credentials.